Practice workbook · Finding where a curve has a given gradient

Setting dy/dx equal to a known gradient (from a parallel line or a normal) to find the points, then using them for tangent constants, lines or lengths.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-CAL-06

1. Question 10 · part e · 7 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part e only.

10 A curve C has equation

$$y = \frac{7-x}{2x+3} \qquad x \neq -\frac{3}{2}$$

(a) Write down an equation of the asymptote to C that is

(i) parallel to the y-axis

(ii) parallel to the x-axis

(2)

(b) Find the coordinates of the points of intersection of C with the coordinate axes.

(2)

(c) Using the axes on the opposite page, sketch C, showing clearly the asymptotes and the coordinates of the points where C crosses the coordinate axes.

(3)

C passes through the point A with coordinates $\left(\frac{1}{2}, \frac{13}{8}\right)$

(d) Show that the gradient of C at A is $-\frac{17}{16}$

(3)

C also passes through the point P

Given that the tangent to C at P is parallel to the tangent to C at A

(e) find an equation of the tangent to C at P
Give your answer in the form $ax + by + c = 0$ where a, b and c are integers.

(7)

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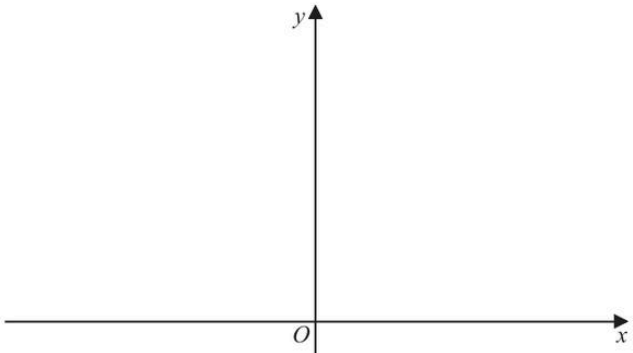
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1. Question 10 · part e · 7 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part e only.

Question 10 continued



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1. Question 10 · part e · 7 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part e only.

Question 10 continued

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1. Question 10 · part e · 7 marks

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Worked-example retry · Attempt part e only.

Question 10 continued

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(Total for Question 10 is 17 marks)

1. Question 10 · part e · 7 marks · Working and answer

Working and answer

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2. Question 10 · part f · 6 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part f only.

10 The curve C has equation

$$y = \frac{5x-2}{ax+b} \qquad x \neq -\frac{b}{a}$$

The asymptote to C that is parallel to the y-axis has equation $x = -\frac{4}{3}$

C crosses the y-axis at the point with coordinates $\left(0, -\frac{1}{4}\right)$

(a) (i) Show that $b = 8$

(ii) Find the value of a

(3)

(b) Write down the equation of the asymptote to C that is parallel to the x-axis.

(1)

(c) Find the coordinates of the point where C crosses the x-axis.

(1)

(d) Using calculus, show that at every point on C, the gradient is positive.

(4)

(e) Using the axes on the next page, sketch C
Clearly label the asymptotes and the coordinates where C crosses the coordinate axes.

(3)

The gradient of C is $\frac{13}{100}$ at the point E and at the point F

(f) Find the length of the line EF, giving your answer to 2 decimal places.

(6)

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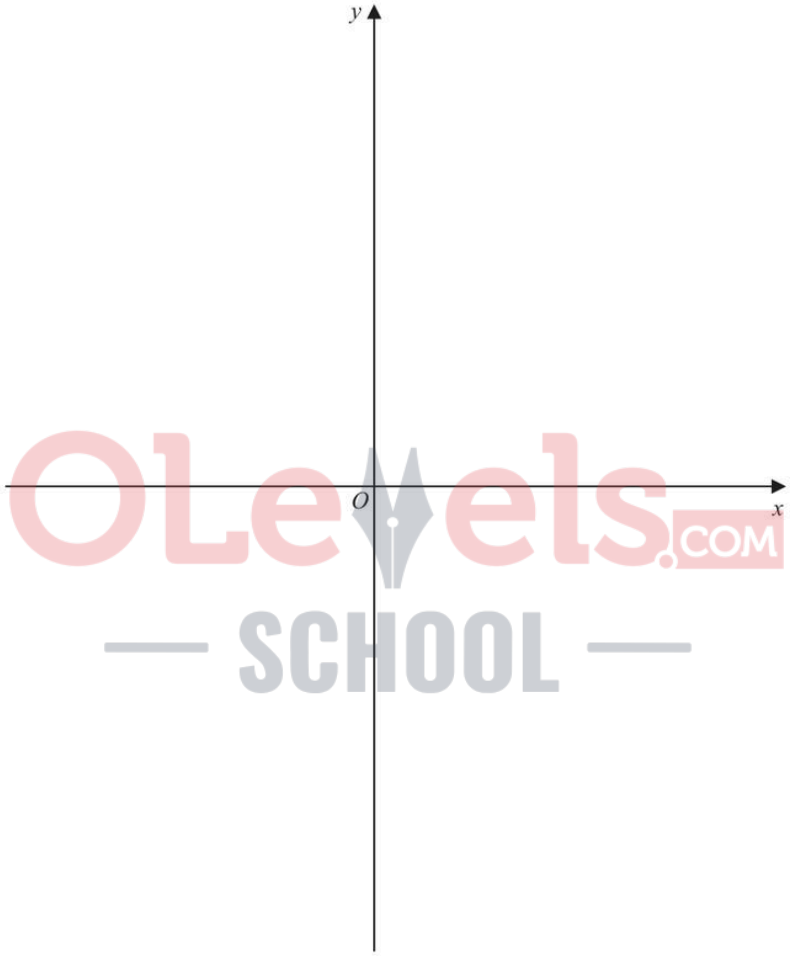
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2. Question 10 · part f · 6 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part f only.

Question 10 continued



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2. Question 10 · part f · 6 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part f only.

Question 10 continued

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2. Question 10 · part f · 6 marks

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Further practice · Attempt part f only.

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Question 10 continued

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(Total for Question 10 is 18 marks)

2. Question 10 · part f · 6 marks · Working and answer

Working and answer

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3. Question 10 · part d.i · 0 marks

2022 JAN · Paper 02

Further practice · Attempt part d.i only.

10 The curve C has equation $y = \dfrac{2x-1}{x+4}$ $\quad x \neq -4$

(a) Write down an equation of the asymptote to C that is

(i) parallel to the x-axis,

(ii) parallel to the y-axis.

(2)

(b) Find the coordinates of the points of intersection of C with the coordinate axes.

(2)

(c) Using the axes on the opposite page, sketch C, showing clearly the asymptotes and the coordinates of the points of intersection of C with the coordinate axes.

(3)

The line with equation $y = x + k_1$ is the tangent to C at the point P

The line with equation $y = x + k_2$ is the tangent to C at the point Q

Given that the x coordinate of P is greater than the x coordinate of Q

(d) using calculus, find the coordinates of

(i) P

(ii) Q

(8)

(e) Hence find the value of

(i) k_1

(ii) k_2

(3)

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3. Question 10 · part d.i · 0 marks · Working and answer

Working and answer

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4. Question 10 · part d.ii · 8 marks

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Further practice · Attempt part d.ii only.

10 The curve C has equation $y = \dfrac{2x-1}{x+4}$ $\quad x \neq -4$

(a) Write down an equation of the asymptote to C that is

(i) parallel to the x-axis,

(ii) parallel to the y-axis.

(2)

(b) Find the coordinates of the points of intersection of C with the coordinate axes.

(2)

(c) Using the axes on the opposite page, sketch C, showing clearly the asymptotes and the coordinates of the points of intersection of C with the coordinate axes.

(3)

The line with equation $y = x + k_1$ is the tangent to C at the point P

The line with equation $y = x + k_2$ is the tangent to C at the point Q

Given that the x coordinate of P is greater than the x coordinate of Q

(d) using calculus, find the coordinates of

(i) P

(ii) Q

(8)

(e) Hence find the value of

(i) k_1

(ii) k_2

(3)

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4. Question 10 · part d.ii · 8 marks · Working and answer

Working and answer

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5. Question 10 · part e.i · 0 marks

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Further practice · Attempt part e.i only.

10 The curve C has equation $y = \dfrac{2x-1}{x+4}$ $\quad x \neq -4$

(a) Write down an equation of the asymptote to C that is

(i) parallel to the x-axis,

(ii) parallel to the y-axis.

(2)

(b) Find the coordinates of the points of intersection of C with the coordinate axes.

(2)

(c) Using the axes on the opposite page, sketch C, showing clearly the asymptotes and the coordinates of the points of intersection of C with the coordinate axes.

(3)

The line with equation $y = x + k_1$ is the tangent to C at the point P

The line with equation $y = x + k_2$ is the tangent to C at the point Q

Given that the x coordinate of P is greater than the x coordinate of Q

(d) using calculus, find the coordinates of

(i) P

(ii) Q

(8)

(e) Hence find the value of

(i) k_1

(ii) k_2

(3)

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5. Question 10 · part e.i · 0 marks · Working and answer

Working and answer

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6. Question 10 · part e.ii · 3 marks

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Further practice · Attempt part e.ii only.

10 The curve C has equation $y = \dfrac{2x-1}{x+4}$ $\quad x \neq -4$

(a) Write down an equation of the asymptote to C that is

(i) parallel to the x-axis,

(ii) parallel to the y-axis.

(2)

(b) Find the coordinates of the points of intersection of C with the coordinate axes.

(2)

(c) Using the axes on the opposite page, sketch C, showing clearly the asymptotes and the coordinates of the points of intersection of C with the coordinate axes.

(3)

The line with equation $y = x + k_1$ is the tangent to C at the point P

The line with equation $y = x + k_2$ is the tangent to C at the point Q

Given that the x coordinate of P is greater than the x coordinate of Q

(d) using calculus, find the coordinates of

(i) P

(ii) Q

(8)

(e) Hence find the value of

(i) k_1

(ii) k_2

(3)

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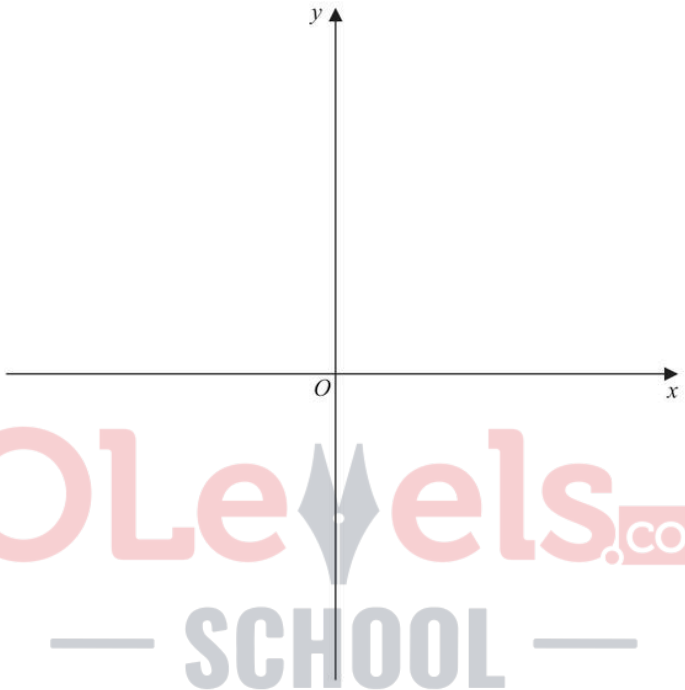
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6. Question 10 · part e.ii · 3 marks

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Further practice · Attempt part e.ii only.

Question 10 continued



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Turn over for spare axes if you need to redraw your sketch.

6. Question 10 · part e.ii · 3 marks

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Further practice · Attempt part e.ii only.

Question 10 continued

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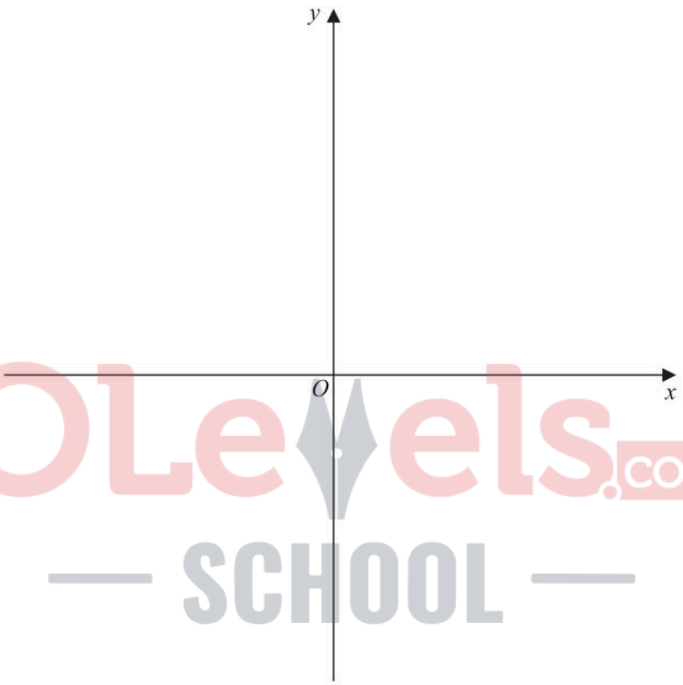
6. Question 10 · part e.ii · 3 marks

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Further practice · Attempt part e.ii only.

Question 10 continued

Only use these axes if you need to redraw your sketch.



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6. Question 10 · part e.ii · 3 marks

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Further practice · Attempt part e.ii only.

Question 10 continued

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(Total for Question 10 is 18 marks)

6. Question 10 · part e.ii · 3 marks · Working and answer

Working and answer

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7. Question 10 · part e · 7 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part e only.

10 A curve C has equation

$$y = \frac{7x - 2}{2x - 3} \qquad x \neq \frac{3}{2}$$

(a) Write down an equation of the asymptote to C that is

(i) parallel to the y-axis,

(ii) parallel to the x-axis.

(2)

(b) Find the coordinates of the points of intersection of C with the coordinate axes.

(2)

(c) Using calculus, show that at every point on the curve, the gradient of C is negative.

(4)

(d) Using the axes on the opposite page, sketch C.
Show clearly and label with their equation any asymptotes and the coordinates of the points of intersection of C with the coordinate axes.

(3)

The straight line l is the normal to C at the point A.
The x coordinate of A is positive and the gradient of l is 17
The line l also intersects C at the point B.

(e) Find the exact coordinates of B.

(7)

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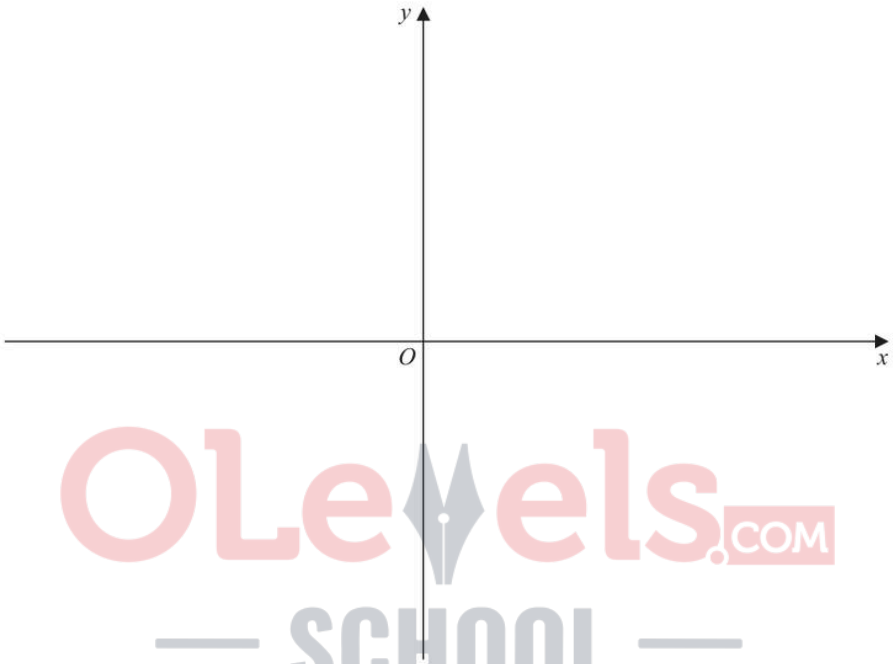
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7. Question 10 · part e · 7 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part e only.

Question 10 continued



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7. Question 10 · part e · 7 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part e only.

Question 10 continued

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7. Question 10 · part e · 7 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part e only.



Question 10 continued

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(Total for Question 10 is 18 marks)

7. Question 10 · part e · 7 marks · Working and answer

Working and answer

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8. Question 9 · part e · 5 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part e only.

9 A curve C has equation $y = \dfrac{3 - 2x}{x + 6}$ where $x \neq -6$

(a) Write down an equation of the asymptote to C that is parallel to the

(i) x-axis (ii) y-axis

(2)

(b) Find the coordinates of the point where C crosses the

(i) x-axis (ii) y-axis

(2)

(c) Using the axes opposite, sketch the graph of C, showing clearly its asymptotes and the coordinates of the points where C crosses the coordinate axes.

(3)

(d) Show that the gradient of the tangent to C is always negative.

(3)

A tangent to C has equation $y = -\dfrac{3}{5}x + k$ where $k > 0$

(e) Find the value of k

(5)

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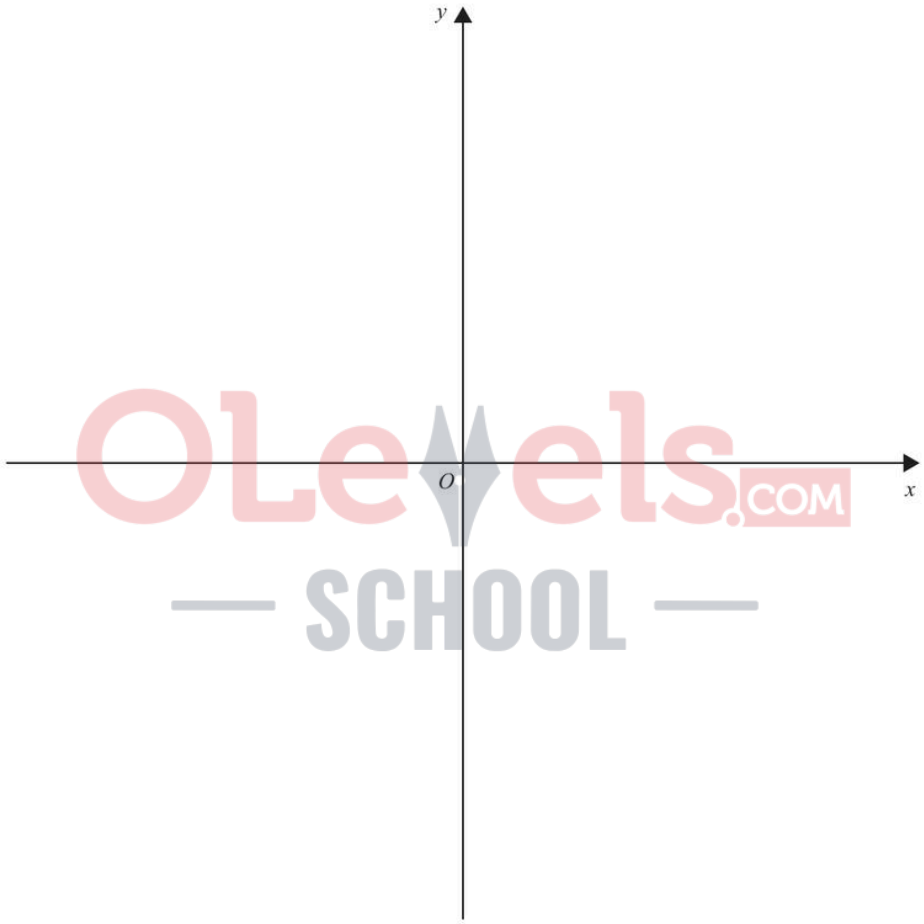
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8. Question 9 · part e · 5 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part e only.

Question 9 continued



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8. Question 9 · part e · 5 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part e only.

Question 9 continued

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8. Question 9 · part e · 5 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part e only.

Question 9 continued

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(Total for Question 9 is 15 marks)

8. Question 9 · part e · 5 marks · Working and answer

Working and answer

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9. Question 10 · part d · 11 marks

2024 NOV · Paper 02

Further practice · Attempt part d only.

10 A curve C has equation

$$y = \frac{5x-2}{3x+2} \qquad x \neq -\frac{2}{3}$$

(a) Find the coordinates of the point where C intersects the

(i) x-axis
(ii) y-axis

(2)

(b) Write down an equation of the asymptote to C that is

(i) parallel to the x-axis
(ii) parallel to the y-axis

(2)

(c) Sketch C on the opposite page.
Show and label the asymptotes and the coordinates of the points where C crosses the coordinate axes.

(3)

Point A lies on C such that the gradient of C at A is parallel to the line with equation $4y - x = 7$

The normal to C at A intersects the x-axis at point D and the y-axis at point E
Given that the x coordinate of A is positive,

(d) find, in its simplified form, the exact length of line DE

(11)

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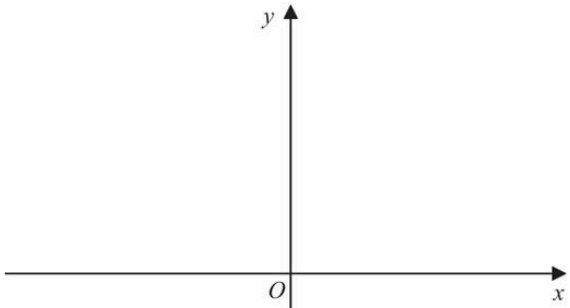
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9. Question 10 · part d · 11 marks

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Further practice · Attempt part d only.

Question 10 continued



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9. Question 10 · part d · 11 marks

2024 NOV · Paper 02

Further practice · Attempt part d only.

Question 10 continued

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9. Question 10 · part d · 11 marks

2024 NOV · Paper 02

Further practice · Attempt part d only.

Question 10 continued

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9. Question 10 · part d · 11 marks

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Further practice · Attempt part d only.

Question 10 continued

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(Total for Question 10 is 18 marks)

9. Question 10 · part d · 11 marks · Working and answer

Working and answer

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10. Question 8 · part a · 5 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

8 $f(x) = 2x^2 - 4x + 9$

The curve C has equation $y = f(x)$
The line l with equation $y - 8x + 9 = 0$ is a tangent to C
The line k is perpendicular to l and is also the tangent to C at the point where $x = p$

(a) Find the value of p

(5)

(b) Show that

(i) $(\alpha + \beta)^3 = \alpha^3 + \beta^3 + 3\alpha\beta(\alpha + \beta)$

(ii) $\alpha^4 + \beta^4 = \left((\alpha + \beta)^2 - 2\alpha\beta\right)^2 - 2(\alpha\beta)^2$

(2)

The quadratic equation $f(x) = 0$ has roots α and β
Without solving the equation and using your results from part (b)

(c) form a quadratic equation with integer coefficients, that has roots

$\alpha^3 - \beta$ and $\beta^3 - \alpha$

(8)

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10. Question 8 · part a · 5 marks · Working and answer

Working and answer

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